



Getting Started

Version 10.0

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Contents

Introduction to the Atari Jaguar Development System.....	3
Setup & Installation	4
Atari Developer Disks.....	4
Installation	4
Configuration	5
Modern Development Tools.....	5
Virtual Jaguar (various authors, currently maintained by Shamus)	5
Hill Software Development Bundle	6
Installation	6
Configuration	6
U-253 Development Tools.....	7
Sound Engine	7
Graphics Converter.....	7
Reboot Development Tools.....	7
Raptor API.....	7
JagStudio.....	7
Jiffi	7
RMAC and RLN	7
Support for the modern development tools.....	7
Running your first program	8
Atari Development System.....	8
SkunkBoard Development System.....	10
How To Run a Cartridge in an Atari Development System	12
How To Boot a CD-ROM in an Atari Development System.....	12
Installation Addendum (Using MS-DOS)	13
Directory (Folder) name examples	13
File name examples	13
Changing Drive and Directory (Folder) locations.....	13
File Rename, File Delete & View Directory Files	13

Introduction to the Atari Jaguar Development System

Atari is proud to introduce the most advanced entertainment console system in the whole industry, the Atari Jaguar. Featuring 64-bit technology and multiple custom RISC processors, the Jaguar has the power to lead interactive entertainment into the 21st century.

The Jaguar development documentation contains information on development hardware, software and describes the development environment. The documentation is not a single document but a collection of documents focusing on different aspects of the Jaguar consoles hardware, development systems, and software programming as this make them easier to maintain.

Included with official development systems and most retail consoles is the Cybermorph game cartridge, the first truly interactive 3D-world game existing at a consumer price level. Cybermorph should give you some idea about the capabilities of the Jaguar. However, while Cybermorph is an impressive game, we would like to emphasise that as one of the earliest Jaguar releases, it only scratches the surface of the machine's capabilities.

Because there are some differences between an official development console and a standard off-the-shelf retail console those with an official development console should refer to the section titled **How To Run A Game Cartridge in an Atari Development System**.

We would like to encourage developers to push the Jaguar system to the limit and design software that takes advantage of the great variety of capabilities offered by the hardware. Push the envelope of reality on the first entertainment system that delivers real *Power Without the Price*TM.

Setup & Installation

There are three basic steps to getting started with your Jaguar development system:

1. Installation
2. Configuration
3. Running your first program

We'll take you through each of these steps from installing the Jaguar development tools and sample programs onto your system to running your first program.

Note: you should read the Installation Addendum at the end of this document before proceeding with your development tools installation.

Atari Developer Disks

Installation

Please read these instructions carefully before trying to install the Jaguar developer tools and a sample code. In order that you have the most up to date instructions, please view the file REA DTHIS.1ST (normally found on Disk 1) prior to attempting installation.

About 15 megabytes of free disk space on your hard disk drive is required for installation. **Note:** You must perform the installation from the MS-DOS command prompt. If you are running Microsoft Windows, please exit to DOS or run the MS-DOS command prompt from within Windows in order to perform the installation.

- 1) Change to the drive and directory where you want to install the files. The installation will automatically create a directory named JAGUAR at this location and install everything into it. (e.g. if you are at D:\ then you'll end up with D:\JAGUAR and it will contain everything. Therefore, you do not need to create a JAGUAR directory yourself.)
- 2) Insert disk #1 into your floppy drive.

To install from a drive A: enter the command `"A:install A:"`

To install from a drive B: enter the command `"B:install B:"`

Type in the commands exactly as shown. Do not use a backslash following the drive letter and colon. **Note:** The drive letter and colon specifies the SOURCE drive, not the destination. The destination is implied by the current drive and a directory when you run the installation.

- 3) The installation process will take several minutes to complete, and you will be prompted to change disks when needed. Simply follow the onscreen instructions. When the installation is complete, you will be returned to the DOS prompt.

Configuration

There are several environment variables used by the Jaguar development tools that need to be set properly before you can do anything. The instructions below apply to an MS-DOS system (with or without Microsoft Windows). If you have a different operating system, you will have to adjust the steps as appropriate.

Add the following lines to your AUTOEXEC.BAT file:

```
set RDBRC=E:\JAGUAR\BIN\RDB.RC
set DBPATH=E:\JAGUAR\BIN
set ALNPATH=E:\JAGUAR\BIN
set MACPATH=E:\JAGUAR\INCLUDE1
set GCC_EXEC_PREFIX=E:/JAGUAR/BIN
set TEMP=C:
set PATH=%PATH%;E:\JAGUAR\BIN
```

You should change “E:\” in the paths above to the drive and directory where the JAGUAR directory is located (this is the path from step #1 of the installation). Having these environment variables set correctly is critical if you want the tools and examples work properly.

You may already have a TEMP environment variable specified in your AUTOEXEC.BAT file. If so, change it say that it specifies just a drive letter and a colon, as shown above. The GNU GCC C compiler may not work properly if you’re TEMP environment variable ends in a backslash.

After you have made changes to your AUTOEXEC.BAT file and saved it back to your hard disk, reboot the machine so they will take effect. For more detailed information about how these environment variables are used, please refer to the documentation for the individual tools.

The RDBPC and DBPATH variables are used by the debugger. The ALNPATH variable is used by the linker. The MACPATH variable is used by the Madmac assembler. The GCC_EXEC_PREFIX variable is used by the GCC C compiler. Note that GCC_EXEC_PREFIX uses a forward slash (“/”) as a path separator instead of a backslash (“\”). Most of the tools also use the PATH and TEMP variables.

Modern Development Tools

Virtual Jaguar (various authors, currently maintained by Shamus)

Nothing is better than testing on real hardware, however if you are unable to locate development hardware such as the Alpine board or SkunkBoard you can get quit good results from using Virtual Jaguar.

Virtual Jaguar is an Atari Jaguar Emulator that runs on Windows, Mac and Linux, and can be downloaded from [here](#).

¹ As of September 26, 1994, the standard system include files have been revised and are now located in the JAGUAR\INCLUDE directory instead of JAGUAR\INC. Some older source code may still require the older versions of the include files, but this should not be a problem with any of the current examples in the developer’s kit.

Hill Software Development Bundle

If you are unable to locate a copy of the original Atari development tools you can download the modern equivalent from either hillSoftware.com (jagdev_102 for Windows and slaxgui for Linux) or the mirror at Harmlesslion.com (Windows only) This toolset includes most of the original Atari tools and examples but replaces the Madmac compiler and ALN linker with SubQMod's updated versions (SMAC and SLN respectively) as well as including the Drivers, JCP up-loader and library for the SkunkBoard.

Installation

About 28 megabytes of free disk space is required for installation.

1. Double click the JAGDEV_XXX application where XXX refers to the version number (102) at the time of writing.
2. Click the next button and enter the installation directory, i.e. D:\Jaguar.

The Jaguar tools and examples source code will be installed, and the environment variables set up automatically. After installation you may find it useful to compile the SkunkBoard library file (Skunk.S) located at **Drive:\Directory\Skunk\Source\SkunkLib** if you do not intend to use Makefiles.

Configuration

Configuration for JAGDEV_XXX is performed automatically by the installer upon installation. However, you may wish to move or copy the SkunkBoard, ALN and SMAC manuals to the "DOCS" folder and the Skunklib.S file to the "lib" folder as it make it easier to remember where they are.

If you are using one of the other Hill Software development systems, you may have to configure them yourself. For example, to access your own files when using the GUI version of LIVE CD you will need to mount you hard drive partition as follows...

1. Go to the system icon / storage media. Find your hard drive and note the device name listed for it when you browse it. Mine says "system:/media/hdb1" in the file browser location window. That is my Windows C drive. Your path may vary.
2. Launch a terminal prompt by clicking on the little black icon on the bottom left of the screen, just to the right of the K icon.
In the terminal window that pops up. I would enter the following (without the quotes) to go to my C:\JAGUAR\3D folder...
"cd /mnt/hdb1/JAGUAR/3D" (note the hdb1 was taken from the file browser location line above (see No.1). The GUI CD mounts partitions it finds and puts them in the /mnt folder under hdXX depending on the partitions. So you might have an hdb1 and hdb2 or hda1 or hda2.

You should note that Linux is case sensitive with filenames.

Once you have changed to your project directory, type "make" and it will attempt to run any makefile you may be using. You may have to edit or tweak the makefile if the paths are different from what the makefile expects or you want to use SMAC/SLN instead of MAC and ALN.

U-253 Development Tools

Sound Engine

This Sound Engine provides simple high-quality audio playback for your Atari Jaguar projects. Simply drop it in, initialise and start making noise, more information and the download is available from [here](#).

Graphics Converter

This Graphics Converter is a tool for converting common graphics formats into raw binary images that can be processed directly by the Atari Jaguar 64bit Multimedia System. More information and the download are available from [here](#)

Reboot Development Tools

Raptor API

The API does "all the stuff" like object calls, screen setup, collision, updates, handling the Object Processor and Blitter, stopping the system crashing, etc.

It is basically the 'guts' of everything required to make the Jaguar do something. It's optimised and efficient, and runs mostly on the GPU, with lesser functions on the 68k. It can be used on its own or linked to anything written in any available language that can link an .obj (.o) file.

The latest supported version of the Raptor API is available [here](#).

JagStudio

JagStudio is the successor the rB+ (Raptor Basic Plus) that now allows you to code for the Jaguar in BASIC (C and Assembler coding support currently in Beta).

JagStudio can be download from [here](#).

Jiffi

Makes bootable Jaguar CD's and can be download from the bottom of the [Reboot download page](#).

RMAC and RLN

Updated and maintained versions of the original Atari Madmac compiler and ALN linker DOS software. They are packed in with JagStudio but can also be used to replace MAC/ALN or SMAC/SLN (as applicable) and can be downloaded from the [official website](#).

The advantages of RMAC/RLN over ALN/MAC are that they run on modern hardware (MAC/ALN requires emulation); support the Jaguar RISC processors (MAC/ALN does not); has a Jaguar Object Processor assembler; and is very fast. Plus, it supports assembling of 6502, 680x0, and DSP 56001 code.

Support for the modern development tools

Support for the Raptor API, JagStudio, U-253 software, HillSoftware development tools, SkunkBoard and Virtual Jaguar can be found on the [Atari Age Jaguar Programming forums](#).

Running your first program

Atari Development System

After you have installed the Jaguar development kit tools and source code, and configured your environment variables, you are ready to compile and run your first program on the Jaguar. Most of the Jaguar development tools are designed to be invoked from the MS-DOS Command prompt. If you are running under Microsoft Windows, you should either exit to DOS or else run the MS-DOS command prompt. If you're running under a different operating system, you should do whatever is required to run MS-DOS programs².

- 1) Change to the JAGUAR\SOURCE\JAGMAND directory³. This directory contains the source code to a Jaguar Mandelbrot fractal program that uses the Jaguar's GPU to calculate the picture of the Mandelbrot set using fast integer arithmetic.
- 2) Type "MAKE" at the DOS command prompt. This will invoke the "MAKE" utility to build the JAGMAND program from the source code. On a DX2/66 machine, this typically takes between 10 and 30 seconds depending on the hard disk and/or network access speed.
- 3) When MAKE is finished, you should have an executable program named JAGMAND.COF. To run it on the Jaguar, we will run the debugger and tell it to load the program into the alpine board.

Before proceeding, let's make sure your PC and Jaguar are properly connected. Your PC should have an 8-bit bidirectional parallel port. (In the event that your PC does not already have such a port, you should install the card supplied with your Jaguar development system. Please see the documentation included with the card.) The Jaguar Alpine board should be plugged into your PC's parallel port using the supplied parallel cable, and the Alpine board itself should be firmly plugged into the cartridge slot of the Jaguar. Make sure that the toggle switch on top of the Alpine board is switched to "Write Enable". If you have not done so already, turn on the Jaguar.

You should see a message similar to:

```
JAGUAR ® Development System
© 1993 Atari Corp.
31 Oct '93
```

on the monitor or television that the Jaguar is connected to. Note that the date shown on your screen and other minor details may be different (particularly if you have a CD-ROM development system). If you do not see this message, you should verify that everything is plugged in correctly.

2 Compatibility has been tested with *Windows v3.1*, *Windows For Workgroups v3.11*, and to a lesser degree with *Windows NT* and the "final beta" version of *Windows 95*. Any compatibility problems with these systems are likely to be related to your specific system setup.
Compatibility with other systems such as OS/2 has not been tested.

3 This was in the JAGUAR\EXAMPLES\JAGMAND directory in older versions of the standard distribution. If you are using this directory, you should search online for a more up to date distribution.

- 4) Enter the command “RDBJAG JAGMAND.COF” at the command prompt. This will load the Jaguar debugger and tell it to load the JAGMAND.COF program. You should see something that looks approximately like this:

```
G:\JAGUAR\EXAMPLES\JAGMAND>rdbjag jagmand.cof
Jaguar Debugger v1.00 PC - May 27 1994
(c)1993 Atari Corporation. PC version by Brainstorm.
Bidirectional parallel port used: LPT1
Jaguar stub (31-Oct-'93) ready and running in ROMULATOR, (NTSC)
COFF program jagmand.cof loaded:
      start   size   end
text  802000   440   802440
data  802440   200   802640
Loaded 304 symbols from COFF program jagmand.cof.
PC: 00802000 SSP: 00000DD2 USP: FFFF7DF7 SR: 2100  SU IPL=1 XC PL NZ VC CC
D 80150014 0080F000 0000000B 0000FFFF FFF70050 FBFF7FFF FFFFFFFF FFFF7FFF
A 0080198A 008006EA 00000E46 00001FFA 00F14000 008015F8 00F00000 00000DD2
00802000> move.l    #$70007,$F0210C G_END
At \JAGUAR\EXAMPLES\JAGMAND\JAGMAND.S:32:
32>      move.l    #$00070007,G_END
Db:
```

If you don't see something essentially like this message, then something may be wrong with your installation, your parallel card may not be recognized as 8-bit bidirectional by RDBJAG, the parallel cable running from your PC's parallel port to the Alpine board isn't plugged in correctly, or there may be something wrong with your Alpine board and/or Jaguar. (Note again that the version numbers and dates may be different on your system.)

- 5) Assuming that things worked as expected in step #4, then type “G” and hit <return> to run the program. The Jaguar should draw an overall view of the Mandelbrot set fractal screen in roughly 8 seconds.

Please note that while the Mandelbrot demo is reasonably speedy, it is not fully optimised and could be made to run even faster. Greater speed could be accomplished by having more work done internally by the GPU, and less by the 68000, and you could also speed things up by having the DSP do some of the calculations. Through these methods, you could probably gain at least a 100% speed increase

Most of the other sample programs supplied with the Jaguar Development System are set up to be compiled and executed in more or less the same way as the JAGMAND demo. Simply move to the directory containing the demo you want, type “MAKE”, and then run the debugger to load the executable into the Jaguar. Note that depending on your system set up, it may be necessary to make slight changes to the MAKEFILE for each demo in order to get things to compile correctly on your system. The **Jaguar Sample Programs** document has more specific information on the various sample programs and how they work.

SkunkBoard Development System

After you have installed the Hill Software development kit tools and source code (JAGDEV_XXX), you should download the latest Skunkboard software from [Harmlesslion](#), now you are ready to compile and run your first program on the Jaguar. Most of the Jaguar development tools are designed to be invoked from the MS-DOS Command prompt. If you are running under Microsoft Windows, you should either exit to DOS or else run the MS-DOS command prompt. If you're running under a different operating system, you should do whatever is required to run MS-DOS programs⁴.

- 1) Change to the JAGUAR\SOURCE\JAGMAND directory. This directory contains the source code to a Jaguar Mandelbrot fractal program that uses the Jaguar's GPU to calculate the picture of the Mandelbrot set using fast integer arithmetic.

Type `rename jagmand.cof mand_old.cof` to rename the existing JAGMAND.COF file or `del jagmand.cof` to delete it.

If you do not do this the MAKE program will report that the COF file is up to date and you will not be able to see what happens when a MAKEFILE is processed.

- 2) Open the MAKEFILE file with a text editor (e.g. Notepad).
As we will be using SMAC and SLN, not Madmac and ALN you should use the "Replace" option located in the File menu to replace "MACFLAGS" with "SMACFLAGS" and "ALNFLAGS" with "SLNFLAGS", although not strictly necessary it probably best to do this to avoid any confusion.

What you absolutely must do is replace "mac" with "smac" and "aln" with "sln".

Finally delete the "-w" flag from the SLNFLAGS list as SLN does not use it, and save the file.

- 3) Type "MAKE" at the DOS command prompt. This will invoke the "MAKE" utility to build the JAGMAND program from the source code. On a DX2/66 machine, this typically takes between 10 and 30 seconds depending on the hard disk and/or network access speed.
- 4) When MAKE is finished, you should have an executable program named JAGMAND.COF. To run it on the Jaguar, we will run the Jaguar Copy Program (JCP) and tell it to load the JAGMAND into the Jaguars RAM.

Before proceeding, let's make sure your PC and Jaguar are properly connected. Your PC should have an USB port, connect the SkunkBoard to you PC using the supplied USB cable, and plug the SkunkBoard board itself firmly plugged into the cartridge slot of the Jaguar.

Once connected you should turn on the Jaguar, you should see a green screen on the monitor or television that the Jaguar is connected to. If you do not see the green screen you should verify that everything is plugged in correctly.

Note: The first time you turn on your Jaguar Windows should recognise the SkunkBoard and install the SkunkBoard drivers, therefore you will have to wait for this process to finish before attempting to issue commands to the SkunkBoard. However, Windows 8 or 10 users should download and run [ZADIG](#), select the "libusb-win32 (v1.2.6.0)" option as the driver for the Skunkboards Cypress EZ-OGT device and ignore sections 4 to 7 of Windows installation instruction in the Skunboard manual.

⁴ Compatibility has been tested with *Windows XP, Windows Vista and Windows 7*. Any compatibility problems with these systems are likely to be related to your specific system setup.

- 5) Enter the command “JCP -f JAGMAND.COF” at the command prompt. This will load the JAGMAND.COF file to the SkunkBoard flash ROM and start it. You should see something that looks approximately like this:

```
D:\JAGUAR\SOURCE\JAGMAND>jcp -f jagmand.cof
jcp v02.03.01 built on Oct 14 2009

COFF File: Skip 168 bytes, base addr is $802000, length is 30610 bytes
Sending...
..
Waiting for erase to complete <about 9s>..
Sending...
Jag accepted start request as $008020000

Finished in 187 millis, 163KB/second.

D:\JAGUAR\SOURCE\JAGMAND>
```

If you don't see something essentially like this message, then something may be wrong with your installation, your SkunkBoard may not be correctly inserted in the Jaguar, the USB cable running from your PC to the SkunkBoard is either defective or isn't plugged in correctly, or there may be something wrong with your SkunkBoard and/or Jaguar.

- 6) Assuming that things worked as expected in step #5, then the Jaguar should draw an overall view of the Mandelbrot set fractal screen in roughly 8 seconds.

Please note that while the Mandelbrot demo is reasonably speedy, it is not fully optimised and could be made to run even faster. Greater speed could be accomplished by having more work done internally by the GPU, and less by the 68000, and you could also speed things up by having the DSP do some of the calculations. Through these methods, you could probably gain at least a 100% speed increase

Most of the other sample programs supplied with the Jaguar Development System are set up to be compiled and executed in more or less the same way as the JAGMAND demo. Simply move to the directory containing the demo you want, type “MAKE”, and then run JCP to load the executable into the Jaguar. Note that depending on your system set up, it may be necessary to make slight changes to the MAKEFILE for each demo in order to get things to compile correctly on your system. The [Jaguar Sample Programs](#) document has more specific information on the various sample programs and how they work.

How To Run a Cartridge in an Atari Development System

- 1) With the Jaguar console turned off, plug in the cartridge in place of the Alpine board.
- 2) Connect a 1K Ohm resistor between pins 4-5 in the STOP cable that normally plugs into the back of the Alpine board. Otherwise the console will not run or might mess up the sound. (Note: Pin 1 on the header of the cable is marked with a small triangle and normally the line leading to pin 1 of the cable is coloured.) Below is a diagram of the header on the Alpine-end of the cable.



- 3) Hold down the 'B' button of the Joypad in controller port #1 and turn on the console's power. Release the 'B' button when you see the Jaguar logo.
- 4) From this point operation is identical to a standard retail console. Hit the 'B' button again to leave the Jaguar logo screen and begin the game.

Note: If you are trying to run a game loaded into a Flash ROM cartridge then you should press the 'C' button instead of 'B' in steps 3 and 4. Note that your development console must have a ROM dated November 1994 or later in order to use Flash ROM cartridges.

How To Boot a CD-ROM in an Atari Development System

If you have a Jaguar CD-ROM development system with a boot ROM installed, you may play standard Jaguar CD-ROM titles. Follow steps 1-4 as shown above, except press button 'C' instead of button 'B'. If there is a Jaguar CD-ROM in the drive, it will be executed. If there is an audio CD in the drive, then the built in **Virtual Light Machine** program will be started.

*If you cannot get the **Virtual Light Machine** program to come up on the screen, your Jaguar CD-ROM unit may not be equipped with the proper boot ROM. Note also that your development console must have a ROM dated November 1994 or later in order to boot from the Jaguar CD-ROM.*

Installation Addendum (Using MS-DOS)

As the Atari development tools/programs were written to run under MS-DOS you should therefore limit your development directory and file names to the old DOS 8+3 format (as per the Atari ST). By doing this you should avoid any compatibility issues with any of the development programs, tools or any of the other development systems mentioned should you decide to change your development system at a later date.

Directory (Folder) name examples

These can be up to eight characters long and must contain only alphanumerical and the underscore (“_”) characters.

JAG JAGUAR JAGUAR_3

are examples of acceptable directory names.

File name examples

These can also be up to eight characters long but with the addition of a maximum three-character file extension and have the same character usage restrictions as directory names.

PROG7.S JAG_PIC3.RGB PROG_9.COF

are examples of acceptable filenames.

Changing Drive and Directory (Folder) locations

For those who are not old enough to remember using MS-DOS within the command prompt window you...

1. Change drives by entering the new drive letter followed by a colon (e.g. D:).
2. Change directories by entering the CD (change directory) command followed by the directory name (e.g. `cd Jaguar\Skunk`).
3. Go back to the previous directory/folder by entering “`CD . .`”.

For example, to change from drive C to the directory “Docs” located in the Jaguar directory on drive D you would enter...

“D:” [ENTER] and then “`cd Jaguar\Docs`” [ENTER]

To return to D from Jaguar\Docs in order to change to another directory/drive enter “`CD . .`”.

File Rename, File Delete & View Directory Files

Rename a file by entering “`rename`” followed by the current filename and then the new filename.

Delete a file by entering “`del`” followed by the filename.

View a list of files and directories contained within the current directory (folder) by entering “`dir /w`”.